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Voltage converter

Abstract

Disclosed herein is a voltage converter including input nodes configured to receive an input voltage, output nodes configured to deliver an output voltage, a rectifying bridge coupled between the input nodes and the output nodes, a capacitor and a resistor series-coupled between the output nodes, and a thyristor coupled between one terminal of the resistor and a given one of the output nodes, wherein the thyristor is configured to allow flow of a positive current from the resistor to the given one of the output nodes. A control input is configured to receive a control signal, wherein the control signal biases a gate of the thyristor to control the flow of current through the thyristor. A transient voltage suppressor circuit is coupled to the gate of the thyristor, configured to activate the thyristor upon exceeding a threshold voltage.

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Background/Summary

PRIORITY CLAIM (1) This is a continuation of U.S. patent application Ser. No. 17/532,717, filed on Nov. 22, 2021, itself claiming the priority benefit of French Application for U.S. Pat. No. 2,012,211, filed on Nov. 26, 2020, the contents of both of which are hereby incorporated by reference in their entirety to the maximum extent allowable by law.

TECHNICAL FIELD

(1) The present disclosure relates generally to electronic devices, and, more specifically, to AC/DC converters. The present disclosure generally applies to any circuit using a rectifying bridge as an AC/DC converter.

BACKGROUND

(2) Many AC/DC converter architectures, based on controllable rectifying elements, for example, thyristors (or SCRs, for Silicon Controlled Rectifiers), or non-controllable rectifying elements, such as diodes assembled as a rectifying bridge, powered with an AC voltage and delivering a DC voltage, are known.

(3) The inrush current, that is, the current peaks which occur on each halfwave of the AC voltage as long as the voltage across a capacitor at the output of the rectifying bridge has not reached a sufficient level, particularly in starting phases, is generally desired to be limited.

(4) There is a need in the art to overcome all or part of the disadvantages of known voltage converters.

SUMMARY

(5) Disclosed herein is a voltage converter including: input nodes configured to receive an input voltage; output nodes configured to deliver an output voltage; a rectifying bridge coupled between the input nodes and the output nodes; a capacitor and a resistor series-coupled between the output nodes; a thyristor coupled between one terminal of the resistor and a given one of the output nodes, wherein the thyristor is configured to allow flow of a positive current from the resistor to the given one of the output nodes; a control input configured to receive a control signal, wherein the control signal biases a gate of the thyristor to control the flow of current through the thyristor; and a transient voltage suppressor circuit coupled to the gate of the thyristor, configured to activate the thyristor upon exceeding a threshold voltage.

(6) The rectifying bridge may include: diodes arranged in parallel branches, each branch having two diodes series-coupled between the output nodes; and inductors coupled between each branch of diodes and respective input nodes.

(7) The control input may receive the control signal as being a positive bias voltage referenced to one of the output nodes.

(8) The thyristor may be a cathode-gate thyristor, and the gate of the cathode-gate thyristor may be configured to receive the control signal to switch the cathode-gate thyristor between an on state and an off state based on a current injected into the gate by the control signal.

(9) The transient voltage suppressor circuit may be configured to form a short circuit between the gate of the thyristor and one of the output nodes when voltage across the thyristor exceeds the threshold voltage, thereby turning on the thyristor to limit the voltage across the resistor during overvoltage conditions.

(10) A diode may be coupled in parallel with the thyristor between a given terminal of the resistor and a given one of the output nodes, a cathode of the diode being connected to the given one of the output nodes and an anode of the diode being connected to the given terminal of the resistor, the diode being configured to allow the flow of a negative current opposite to the positive current allowed by the thyristor.

(11) The resistor may be configured to attenuate a current peak appearing at startup of the voltage converter by providing a path for the current during a period when the thyristor is turned off.

(12) A diode may be coupled in parallel and head-to-tail with the thyristor, configured to allow flow of a negative current opposite to the positive current allowed by the thyristor.

(13) The diode may be coupled in reverse, such that it is non-conductive of the positive current flowing from the resistor, thereby providing that the positive current flows through the resistor during startup of the voltage converter.

(14) Also disclosed herein is a method of converting voltage using a voltage converter, the method including: receiving an input voltage at input nodes of the voltage converter; rectifying the input voltage using a rectifying bridge to produce an output voltage; delivering the output voltage between output nodes of the voltage converter; directing a portion of the output voltage through a series arrangement of a capacitor and a resistor; controlling a flow of a positive current through a thyristor coupled between one terminal of the resistor and a given one of the output nodes by applying a control signal to a gate of the thyristor; and activating the thyristor using a transient voltage suppressor circuit when a voltage threshold is exceeded to manage voltage peaks.

(15) The rectifying may include using diodes arranged in parallel branches with each branch having two diodes series-coupled between the output nodes.

(16) The control signal may be a positive bias voltage referenced to the output node.

(17) Controlling the flow of the positive current through the thyristor may further include injecting a control current into the gate of the thyristor to switch the thyristor between an on state and an off state.

(18) The activating of the thyristor may include detecting a voltage across the thyristor exceeding the voltage threshold forming a short circuit between the gate of the thyristor and the output node to turn on the thyristor.

(19) The method may further include allowing a negative current to flow through a diode coupled in parallel with the thyristor, wherein the diode is non-conductive of the positive current so that the positive current flows through the resistor during startup of the voltage converter.

(20) The directing of the portion of the output voltage through the resistor may include attenuating a current peak at the starting of the converter by providing a path for the current through the resistor while the thyristor is turned off.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing features and advantages, as well as others, will be described in detail in the following description of specific embodiments given by way of illustration and not limitation with reference to the accompanying drawings, in which:

(2) FIG. 1 schematically shows an embodiment of a voltage converter disclosed herein;

(3) FIG. 2 shows in further detail an embodiment of a voltage converter disclosed herein; and

(4) FIG. 3 shows in further detail another embodiment of a voltage converter disclosed herein.

DETAILED DESCRIPTION

(5) Like features have been designated by like references in the various figures. In particular, the structural and/or functional features that are common among the various embodiments may have the same references and may dispose identical structural, dimensional and material properties.

(6) For the sake of clarity, the steps and elements that are useful for an understanding of the embodiments described herein have been illustrated and described in detail. In particular, the various possible applications of the embodiments of voltage converters are not detailed.

(7) Unless indicated otherwise, when reference is made to two elements connected together, this signifies a direct connection without any intermediate elements other than conductors, and when reference is made to two elements coupled together, this signifies that these two elements can be connected or they can be coupled via one or more other elements.

(8) In the following disclosure, unless otherwise specified, when reference is made to absolute positional qualifiers, such as the terms “front”, “back”, “top”, “bottom”, “left”, “right”, etc., or to

relative positional qualifiers, such as the terms “above”, “below”, “higher”, “lower”, etc., or to qualifiers of orientation, such as “horizontal”, “vertical”, etc., reference is made to the orientation shown in the figures.

(9) Unless specified otherwise, the expressions “around”, “approximately”, “substantially” and “in the order of” signify within 10%, and preferably within 5%.

(10) FIG. 1 schematically shows an embodiment of a voltage converter **10**.

(11) Voltage converter **10** receives, as an input, a voltage V_{in} . In other words, converter **10** comprises two input nodes **12** and **14**, with voltage V_{in} being delivered between nodes **12** and **14**. Input nodes **12** and **14** are, for example, coupled across an AC voltage source (not shown) which may be, for example, an electric power distribution system.

(12) Converter **10** outputs an output voltage V_{out} . Voltage V_{out} is, for example, a DC voltage. Voltage V_{out} is delivered between two nodes **16** and **18**. Node **18** corresponds to a reference potential, for example, ground. Voltage V_{out} is referenced to the reference potential on node **18**.

(13) Converter **10** comprises a circuit **20** receiving voltage V_{in} and delivering voltage V_{out} . Circuit **20** is coupled, preferably connected, to input nodes **12** and **14** and to output nodes **16** and **18**. Circuit **20** comprises a rectifying bridge, details of which are not shown in FIG. 1.

(14) Converter **10** comprises a capacitor **22** and a resistor (R) **24** series-coupled between output nodes **16** and **18**. More particularly, one of the terminals of capacitor **22** is coupled, preferably connected, to node **16**. A second terminal of the capacitor is coupled, preferably connected, to a node **26**. A terminal of resistor **24** is coupled, preferably connected, to node **26**. The other terminal of resistor **24** is coupled, preferably connected, to node **18**.

(15) Converter **10** further comprises a circuit **28**. A terminal of circuit **28** is coupled, preferably connected, to node **26**. Another terminal of circuit **28** is coupled, preferably connected, to node **18**. Circuit **28** comprises a control input or terminal. The control input receives a control signal $Comm$, more particularly a positive bias voltage referenced to the voltage on node **18**, in other words, referenced to the same voltage as the output voltage (that is, for example, referenced to ground), and a control current.

(16) Circuit **28** corresponds to a bidirectional switch. In other words, a current may flow through circuit **28** in one direction or in the opposite direction. In other words, a positive current may flow from node **26** to node **18** and a negative current can flow from node **18** to node **26**. At least one of the directions, namely the direction from node **26** to node **18** corresponding to a positive current, may be controlled by control signal $Comm$. Thus, preferably, circuit **28** conducts a positive current from node **26** to node **18** for one or a plurality of first values of the control current and blocks (that is, prevents the flowing of) a negative current for one or a plurality of second values of the control current.

(17) According to an embodiment, the negative current flows whatever is a value of control signal $Comm$.

(18) According to another embodiment, the negative current is conducted by circuit **28** for a signal $Comm$ having a value among the first value(s), and is blocked by circuit **28** for a signal $Comm$ having a value among the second value(s). In other words, the first value(s) enable the current to flow in both directions, and the second value(s) block the flowing of the current in both directions.

(19) At the starting of operation of the converter, a current peak generally appears in the current flowing through capacitor **22**. Such a current peak might cause damage to the components. To limit such a current peak, it is desired to run the current through resistor **24** to attenuate the peak without attenuating the current outside of the peak. Thus, circuit **28** blocks the flowing of the positive current from node **26** to node **18** through the branch comprising circuit **28**, and ensures that the current flows through resistor **24**. In other words, during the period likely to comprise a current peak, that is, at the starting of the converter, the control current has a value such that a positive current is blocked between the input and the output of circuit **28** and such that said positive current flows through the resistor.

(20) When such a current peak appears at the starting of the converter, the current flows from node **16** to node **18**. It is thus generally not useless to block the negative current flowing through circuit **28**.

(21) FIG. 2 shows in further detail an embodiment of a voltage converter **50**. Converter **50** is an embodiment of the converter **10** of FIG. 1. Thus, converter **50** comprises capacitor **22**, circuit **28**, and circuit **20**, coupled as described in relation with FIG. 1.

(22) Circuit **20** comprises a rectifying bridge. In the example of FIG. 2, the rectifying bridge is a diode rectifying bridge. The rectifying bridge thus comprises four diodes **52**, **54**, **56**, and **58**.

(23) Diodes **52** and **56** are series-coupled between node **16** and node **18**, that is, between the output nodes, that is, between the nodes of application of output voltage V_{out} . More particularly, diode **52** is coupled between node **16** and a node **60** and diode **56** is coupled between node **60** and node **18**. In other words, the cathode of diode **52** is coupled, preferably connected, to node **16** and the anode of diode **52** is coupled, preferably connected, to node **60**. Further, the cathode of diode **56** is coupled, preferably connected, to node **60** and the anode of diode **56** is coupled, preferably connected, to node **18**.

(24) Diodes **54** and **58** are series-coupled between node **16** and node **18**, that is, between the output nodes, that is, between the nodes of application of output voltage V_{out} . More particularly, diode **54** is coupled between node **16** and a node **62**, and diode **58** is coupled between node **62** and node **18**. In other words, the cathode of diode **54** is coupled, preferably connected, to node **16** and the anode of diode **54** is coupled, preferably connected, to node **62**. Further, the cathode of diode **58** is coupled, preferably connected, to node **62** and the anode of diode **58** is coupled, preferably connected, to node **18**.

(25) Thus, the rectifying bridge comprises two branches coupled in parallel, each branch comprising two diodes coupled in series, diodes **52** and **56** or diodes **54** and **58**. The diodes of a same branch are series-coupled so that the cathode of one of the two diodes is coupled, preferably connected, to the anode of the other diode.

(26) As a variant, at least some of diodes **52**, **54**, **56**, and **58** may be replaced with another electronic component enabling forming of a rectifying bridge. For example, at least one of diodes **52**, **54**, **56**, and **58**, for example, two of diodes **52**, **54**, **56**, and **58**, may be replaced with controllable elements, for example, thyristors or transistors.

(27) Circuit **20**, for example, comprises inductances **64** and **66**. Inductance **64** is, for example, coupled between node **60** and input node **12**. Inductance **66** is, for example, coupled between node **62** and input node **14**. More particularly, a terminal of inductance **64** is coupled, preferably connected, to node **60** and the other terminal of inductance **64** is coupled, preferably connected, to node **12**. Similarly, a terminal of inductance **66** is coupled, preferably connected, to node **62** and the other terminal of inductance **66** is coupled, preferably connected, to node **14**.

(28) Circuit **28** comprises a diode **68**. Diode **68** is coupled between node **26** and node **18**. Diode **68** is coupled in reverse. In other words, diode **68** is coupled to allow the flowing of a negative current from node **18** to node **26**. In other words, the cathode of diode **68** is coupled, preferably connected, to node **26**. The anode of diode **68** is coupled, preferably connected, to node **18**. Diode **68** is thus coupled in parallel to resistor **24**.

(29) Circuit **28** comprises a thyristor **70**. Thyristor **70** is coupled between node **26** and node **18**. Thyristor **70** is coupled to allow the flowing of a positive current from node **26** to node **18**. In other words, the cathode of thyristor **70** is coupled, preferably connected, to node **18**. The anode of thyristor **70** is coupled, preferably connected, to node **26**. Thyristor **70** is thus coupled in parallel to resistor **24**. Thyristor **70** is thus coupled in parallel and head-to-tail with diode **68**.

(30) Thyristor **70** is, for example, a cathode-gate thyristor. The thyristor receives on its gate a voltage referenced to node **18**. The thyristor gate receives the bias voltage of control signal Comm. The thyristor's gate is thus biased by a positive voltage referenced to node **18**. A control current is injected into the gate to determine the on or off state of the thyristor.

(31) At the starting of the converter, a positive current flows from node **26** to node **18**. The control current has, at the starting, a value enabling to turn on thyristor **70**. Further, diode **68** is non-conductive of the current flowing from node **26** to node **18**. The current thus flows through resistor **24**, which enables attenuation of the current peak appearing at the starting.

(32) After the starting, that is, after the current peak generated at the starting, the control current takes a value turning on thyristor **70**. Thus, during the operation, that is, during the steady state of the converter, the current at least partially, preferably mostly, flows through diode **68** or thyristor **70**, according to the direction of the current.

(33) Preferably, circuit **28** comprises a transient voltage suppressor, TSV, circuit **72**. Circuit **72** enables avoiding damage that might be caused to the converter by a transient voltage or overvoltage when the system is off, in other words when the capacitor **22** is discharged and the thyristor **70** is blocked, meaning that the thyristor **70** is not receiving a control signal corresponding to an on state. For example, during an overvoltage caused by lightning, a strong current flows through the capacitor **22** and through the resistor **24**. Under the effect of this current, the voltage across the terminals of the resistor **24** increases and reaches the threshold voltage of circuit **72**. Once this threshold voltage reached, a current flows from node **23** to node **18**, through circuit **72**, by the triggering of the thyristor **70** and causing of the change of the thyristor to an on state. The thyristor **70**, by becoming on, short-circuits the resistor **24** and stops the voltage to reach a critical value for the converter. For example, circuit **72** is a transil diode, for example a unidirectional transil diode. Preferably, circuit **72** is coupled between node **26** and the gate of thyristor **70**. A terminal of circuit **72** is for example coupled, preferably connected, to node **26** and another terminal of circuit **72** is for example coupled, preferably connected, to node **18**.

(34) Preferably, circuit **72** is configured to form a short-circuit between the gate of thyristor **70** and node **26** if the voltage across thyristor **70** is, for example, greater than 50 V, preferably greater than 100 V, preferably greater than 200 V. Thus, when the voltage across thyristor **70** is greater than the selected threshold, the thyristor receives at its gate the voltage at node **26** and turns on, allowing the dissipation of the current peak in the thyristor, while limiting the voltage to which a circuit powered by the voltage V_{out} is subjected.

(35) FIG. 3 shows in further detail an embodiment of a voltage converter **80**. Converter **80** is an embodiment of the converter **10** of FIG. 1. Thus, converter **80** comprises capacitor **22**, circuit **28**, and circuit **20**, coupled as described in relation with FIG. 1. Circuit **20** comprises, in the example of FIG. 3, the same elements as the circuit **20** described in relation with FIG. 2, coupled in the same way. The circuit **20** of FIG. 3 is thus identical to the circuit **20** of FIG. 2.

(36) Converter **80** of FIG. 3 differs from the converter **50** of FIG. 2 by the composition of circuit **28**. The circuit **28** of FIG. 3 comprises a triac **82** coupled between node **26** and node **18**. In other words, a terminal of triac **82**, preferably the first anode, that is, the anode on the gate side, is coupled, preferably connected, to node **26** and another terminal of triac **82**, for example, the second anode, is coupled, preferably connected, to node **18**.

(37) Preferably, triac **82** is capable of operating in the first and third quadrants Q1 and Q3. First quadrant Q1 indicates an operating state where the current flows from node **18** to node **26**, the voltage on node **18** is greater than the voltage on node **26**, and the current in the gate of triac **82** flows towards triac **82**. Third quadrant Q3 indicates an operating state where the current flows from node **26** to node **18**, the voltage on node **26** is greater than the voltage on node **18**, and the current in the gate of triac **82** flows from triac **82**.

(38) Triac **82** is preferably capable of withstanding high currents, for example, powers in the range from 500 W to 10 kW under a voltage of 230 V.

(39) Circuit **28** further comprises a triac **84** coupled between the gate of triac **82** and node **18**. In other words, a terminal of triac **84**, preferably, the first anode, is coupled, preferably connected, to node **18** and another terminal of triac **84**, for example, the second anode, is coupled, preferably connected, to the gate of triac **82**. Triac **84** receives the control current $Comm$ on its gate to control

triac **84**. Further, the gate of triac **84** is biased by the bias voltage. Thus, the gate of triac **84** is biased to a positive voltage referenced to node **18**.

(40) Preferably, triac **84** is capable of operating in the first and fourth quadrants Q1 and Q4. Fourth quadrant Q4 indicates an operating state where the current flows from node **26** to node **18**, the voltage on node **26** is greater than the voltage on node **18**, and the current in the gate of triac **84** flows towards triac **84**.

(41) At the starting of the converter, a positive current flows from node **26** to node **18**. The control current has, at the starting, a value, for example a null value, enabling to turn off triac **84**. Triac **82** is thus off. The current thus flows through resistor **24**, which enables to decrease the current peak appearing at the starting.

(42) After the starting, that is, after the current peak generated at the starting, the control current takes a value turning on triac **84**, and thus turning on triac **82**. Thus, during the operation, that is, during the steady state of the converter, the current at least partially flows, preferably mostly, through triac **82**.

(43) Preferably, circuit **28** comprises a transient voltage suppressor, TSV, circuit **86**. Circuit **86** enables avoiding damage that could be caused to the converter by an overcurrent when the system is off, for example, an over current caused by lightning. For example, circuit **86** is a transil diode. Preferably, circuit **86** is coupled between node **18** and the gate of triac **82**. A terminal of circuit **86** is for example coupled, preferably connected, to node **18** and another terminal of circuit **86** is for example coupled, preferably connected, to the gate of triac **82**.

(44) Preferably, circuit **86** is configured to form a short-circuit between the gate of triac **82** and node **18** if the voltage across triac **82** is for example greater than 50 V, preferably greater than 100 V, preferably greater than 200 V. Thus, when the voltage across triac **82** is greater than the selected threshold, triac **82** receives on its gate the voltage on node **18** and turns on, allowing the dissipation of the current peak in triac **82**.

(45) An advantage of the described embodiments is that they enable attenuation of current peaks in capacitor **22** at the starting of the converter.

(46) An advantage of the described embodiments is that they enable use of a control signal of circuit **28** referenced to the same node as the other voltage already present in the circuit, for example, at the same node as voltage Vout. It is thus not necessary to supply a different auxiliary power and it is possible to use a voltage already present in the device.

(47) Various embodiments and variants have been described. Those skilled in the art will understand that certain features of these various embodiments and variants may be combined, and other variants will occur to those skilled in the art.

(48) Finally, the practical implementation of the described embodiments and variations is within the abilities of those skilled in the art based on the functional indications given hereabove.

Claims

1. A voltage converter, comprising: input nodes configured to receive an input voltage; output nodes configured to deliver an output voltage; a rectifying bridge coupled between the input nodes and the output nodes; a capacitor and a resistor series-coupled between the output nodes; a thyristor having an anode coupled to one terminal of the resistor and a cathode coupled to a given one of the output nodes, wherein the thyristor is configured to allow flow of a positive current from the resistor to the given one of the output nodes; a control input configured to receive a control signal, wherein the control signal biases a gate of the thyristor to control the flow of current through the thyristor; and a transient voltage suppressor circuit coupled between the anode and the gate of the thyristor, configured to activate the thyristor when a voltage across the anode and cathode of the thyristor exceeds a threshold voltage by creating a short circuit path between the gate and the output node coupled to the cathode of the thyristor.

2. The voltage converter of claim 1, wherein the rectifying bridge comprises: diodes arranged in parallel branches, each branch having two diodes series-coupled between the output nodes; and inductors coupled between each branch of diodes and respective input nodes.
3. The voltage converter of claim 1, wherein the control input receives the control signal as being a positive bias voltage referenced to one of the output nodes.
4. The voltage converter of claim 1, wherein the thyristor is a cathode-gate thyristor, and wherein the gate of the cathode-gate thyristor is configured to receive the control signal to switch the cathode-gate thyristor between an on state and an off state based on a current injected into the gate by the control signal.
5. The voltage converter of claim 1, wherein the turning on of the thyristor by the transient voltage suppressor circuit serves to limit the voltage across the resistor during overvoltage conditions.
6. The voltage converter of claim 1, further comprising a diode coupled in parallel with the thyristor between a given terminal of the resistor and a given one of the output nodes, wherein a cathode of the diode is connected to the given one of the output nodes and an anode of the diode is connected to the given terminal of the resistor, configured to allow the flow of a negative current opposite to the positive current allowed by the thyristor.
7. The voltage converter of claim 1, wherein the resistor is configured to attenuate a current peak appearing at startup of the voltage converter by providing a path for the current during a period when the thyristor is turned off.
8. The voltage converter of claim 1, further comprising a diode coupled in parallel and head-to-tail with the thyristor, configured to allow flow of a negative current opposite to the positive current allowed by the thyristor.
9. The voltage converter of claim 8, wherein the diode is coupled in reverse, such that it is non-conductive of the positive current flowing from the resistor, thereby providing that the positive current flows through the resistor during startup of the voltage converter.
10. The voltage converter of claim 1, wherein the transient voltage suppressor circuit comprises a transil diode configured to form the short circuit between the gate of the thyristor and the output node coupled to the cathode of the thyristor when the voltage across the anode and the cathode exceeds the threshold voltage.
11. A method of converting voltage using a voltage converter, the method comprising: receiving an input voltage at input nodes of the voltage converter; rectifying the input voltage using a rectifying bridge to produce an output voltage; delivering the output voltage between output nodes of the voltage converter; directing a portion of the output voltage through a series arrangement of a capacitor and a resistor; controlling a flow of a positive current through a thyristor coupled between one terminal of the resistor and a given one of the output nodes by applying a control signal to a gate of the thyristor; and activating the thyristor using a transient voltage suppressor circuit coupled between an anode and a gate of the thyristor when a voltage between the anode and the cathode of the thyristor exceeds a voltage threshold by creating a short circuit path between the gate and a given one of the output nodes that is coupled to the cathode of the thyristor.
12. The method of claim 11, wherein the rectifying includes using diodes arranged in parallel branches with each branch having two diodes series-coupled between the output nodes.
13. The method of claim 11, wherein the control signal comprises a positive bias voltage referenced to the output node.
14. The method of claim 11, wherein controlling the flow of the positive current through the thyristor further comprises injecting a control current into the gate of the thyristor to switch the thyristor between an on state and an off state.
15. The method of claim 11, further comprising allowing a negative current to flow through a diode coupled in parallel with the thyristor, wherein the diode is non-conductive of the positive current so that the positive current flows through the resistor during startup of the voltage converter.
16. The method of claim 11, wherein the directing of the portion of the output voltage through the

resistor includes attenuating a current peak at the starting of the converter by providing a path for the current through the resistor while the thyristor is turned off.
